

Math Sec 1

1. If the image of the point $P(1, 2, a)$ in the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z}{2}$ is $Q(5, b, c)$, then $a^2 + b^2 + c^2$ is equal to
(A) 264 (B) 293 (C) 298 (D) 283
2. Let $\vec{AB} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{AD} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \lambda \in \mathbb{R}$. Let the projection of the vector $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ on the diagonal $\vec{AC}$ of the parallelogram ABCD be of length one unit. If α, β , where $\alpha > \beta$, be the roots of the equation $\lambda^2 x^2 - 6\lambda x + 5 = 0$, then $2\alpha - \beta$ is equal to:
(A) 3 (B) 6 (C) 1 (D) 4
3. If the sum of the first four terms of an A.P. is 6 and the sum of its first six terms is 4, then the sum of its first twelve terms is
(A) -22 (B) -26 (C) -20 (D) -24
4. If the line $\alpha x + 2y = 1$, where $\alpha \in \mathbb{R}$, does not meet the hyperbola $x^2 - 9y^2 = 9$, then a possible value of α is:
(A) 0.7 (B) 0.6 (C) 0.5 (D) 0.8
5. If the chord joining the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ on the parabola $y^2 = 12x$ subtends a right angle at the vertex of the parabola, then $x_1 x_2 - y_1 y_2$ is equal to
(A) 292 (B) 288 (C) 280 (D) 284
6. Let the line $x = -1$ divide the area of the region $\{(x, y) : 1 + x^2 \leq y \leq 3 - x\}$ in the ratio $m : n$, $\gcd(m, n) = 1$. Then $m + n$ is equal to
(A) 26 (B) 25 (C) 27 (D) 28
7. Let the relation R on the set $M = \{1, 2, 3, \dots, 16\}$ be given by $R = \{(x, y) : 4y = 5x - 3, x, y \in M\}$. Then the minimum number of elements required to be added in R , in order to make the relation symmetric, is equal to
(A) 4 (B) 1 (C) 3 (D) 2
8. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{|x|+4} \right) dx$, where $[\cdot]$ denotes the greatest integer function, is
(A) $\frac{1}{60}(\pi - 7)$ (B) $\frac{1}{60}(21\pi - 1)$
(C) $\frac{7}{60}(3\pi - 1)$ (D) $\frac{7}{60}(\pi - 3)$
9. If a random variable x has the probability distribution

x	0	1	2	3	4	5	6	7
$P(x)$	0	2	k	3	$2k^2$	$2k$	$k^2 +$	$7k^2$
		k		k			k	

then $P(3 < x \leq 6)$ is equal to
(A) 0.33 (B) 0.34 (C) 0.22 (D) 0.64
10. Two distinct numbers a and b are selected at random from $1, 2, 3, \dots, 50$. The probability that their product ab is divisible by 3 is
(A) $\frac{272}{1225}$ (B) $\frac{8}{25}$
(C) $\frac{664}{1225}$ (D) $\frac{561}{1225}$
11. Let $P(\alpha, \beta, \gamma)$ be the point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at a distance $4\sqrt{14}$ from the point $(1, -1, 0)$ and nearer to the origin. Then the shortest distance, between the lines $\frac{x-\alpha}{1} = \frac{y-\beta}{2} = \frac{z-\gamma}{3}$ and $\frac{x+5}{2} = \frac{y-10}{1} = \frac{z-3}{1}$, is equal to
(A) $4\sqrt{\frac{7}{5}}$ (B) $4\sqrt{\frac{5}{7}}$
(C) $7\sqrt{\frac{5}{4}}$ (D) $2\sqrt{\frac{7}{4}}$
12. If $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, then the determinant of the matrix $(A^{2025} - 3A^{2024} + A^{2023})$ is
(A) 28 (B) 24 (C) 16 (D) 12
13. Let the solution curve of the differential equation $x dy - y dx = \sqrt{x^2 + y^2} dx, x > 0, y(1) = 0$, be $y = y(x)$. Then $y(3)$ is equal to
(A) 6 (B) 1 (C) 2 (D) 4
14. Let $f(x) = x^{2025} - x^{2000}, x \in [0, 1]$ and the minimum value of the function $f(x)$ in the interval $[0, 1]$ be $(80)^{80}(n)^{-81}$. Then n is equal to
(A) -41 (B) -81 (C) -80 (D) -40
15. Let the set of all values of r , for which the circles $(x+1)^2 + (y+4)^2 = r^2$ and $x^2 + y^2 - 4x - 2y - 4 = 0$ intersect at two distinct points be the interval (α, β) . Then $\alpha\beta$ is equal to
(A) 25 (B) 21 (C) 24 (D) 20
16. Let $f : [1, \infty) \rightarrow \mathbb{R}$ be a differentiable function. If $6 \int_1^x f(t) dt = 3xf(x) + x^3 - 4$ for all $x \geq 1$, then the value of $f(2) - f(3)$ is
(A) 3 (B) 4 (C) -3 (D) -4
17. If the domain of the function $f(x) = \sin^{-1}\left(\frac{5-x}{3+2x}\right) + \frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma) - \{\delta\}$, then $6(\alpha + \beta + \gamma + \delta)$ is equal to
(A) 70 (B) 67 (C) 68 (D) 66
18. The coefficient of x^{48} in $(1+x) + 2(1+x)^2 + 3(1+x)^3 + \dots + 100(1+x)^{100}$ is equal to
(A) $100 \cdot {}^{100}C_{49} - {}^{100}C_{50}$

(B) $^{100}C_{50} + ^{101}C_{49}$

(C) $100 \cdot ^{101}C_{49} - ^{101}C_{50}$

(D) $100 \cdot ^{100}C_{49} - ^{100}C_{48}$

19. The number of solutions of $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$, is equal to

(A) 2 (B) 3 (C) 1 (D) 0

20. The number of distinct real solutions of the equation $x|x+4| + 3|x+2| + 10 = 0$ is

(A) 3 (B) 1 (C) 2 (D) 0

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21. Let A be a 3×3 matrix such that $A + A^T = O$. If

$$A \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix},$$

$$A^2 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 19 \\ -24 \end{bmatrix}$$

and $\det(\text{adj}(2 \text{adj}(A + I))) = (2)^\alpha \cdot (3)^\beta \cdot (11)^\gamma$,

α, β, γ are non-negative integers, then $\alpha + \beta + \gamma$ is equal to _____

22. Let ABC be a triangle. Consider four points p_1, p_2, p_3, p_4 on the side AB , five points p_5, p_6, p_7, p_8, p_9 on the side BC , and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC . None of these points is a vertex of the triangle ABC . Then the total number of pentagons that can be formed by taking all the vertices from the points $p_1, p_2, \dots, p_{13}$ is _____.

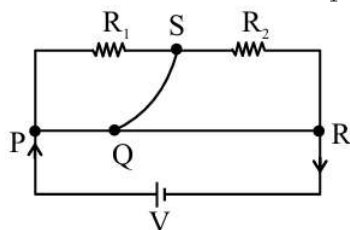
23. Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $\beta = \frac{-1-i\sqrt{3}}{2}$, $i = \sqrt{-1}$. If $(7 - 7\alpha + 9\beta)^{20} + (9 + 7\alpha - 7\beta)^{20} + (-7 + 9\alpha + 7\beta)^{20} + (14)$ then m is _____.

24. If $\frac{\cos^2 48^\circ - \sin^2 12^\circ}{\sin^2 24^\circ - \sin^2 6^\circ} = \frac{\alpha + \beta\sqrt{5}}{2}$, where $\alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to

25. If $\int (\sin x)^{-\frac{11}{2}} (\cos x)^{-\frac{5}{2}} dx = -\frac{p_1}{q_1} (\cot x)^{\frac{9}{2}} - \frac{p_2}{q_2} (\cot x)^{\frac{5}{2}} - \frac{p_3}{q_3} (\cot x)^{\frac{1}{2}} + \frac{p_4}{q_4} (\cot x)^{-\frac{3}{2}} + C$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{15p_1 p_2 p_3 p_4}{q_1 q_2 q_3 q_4}$ is equal to _____

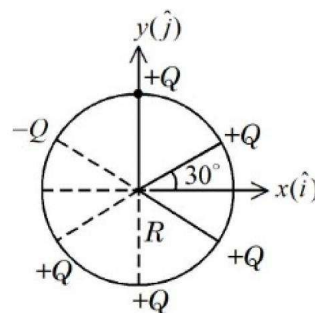
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26. A meter bridge with two resistances R_1 and R_2 as shown in figure was balanced (null point) at 40 cm from the point P . The null point changed to 50 cm from the point P , when 16Ω resistance is connected in parallel to R_2 . The values of resistances R_1 and R_2 are _____.



- (A) $R_2 = 4\Omega, R_1 = \frac{4}{3}\Omega$
 (B) $R_2 = 16\Omega, R_1 = \frac{16}{3}\Omega$
 (C) $R_2 = 8\Omega, R_1 = \frac{16}{3}\Omega$
 (D) $R_2 = 12\Omega, R_1 = \frac{12}{3}\Omega$

27. Six point charges are kept 60° apart from each other on the circumference of a circle of radius R as shown in figure. The net electric field at the center of the circle is _____ . (ϵ_0 is permittivity of free space)



- (A) $-\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$
 (B) $-(\frac{5Q}{8\pi\epsilon_0 R^2})(\hat{i} - 3\hat{j})$
 (C) $\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$
 (D) $-\frac{5Q}{8\pi\epsilon_0 R^2}(\hat{i} + \sqrt{3}\hat{j})$

28. Rods x and y of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points A and F are maintained at 100°C and 40°C respectively. Given the thermal conductivity of rod x is three times of that of rod y , the temperature at junction points B and E are (close to):